



Abd Alrahman Ahmed Mousa Mohamed

Computer Science · Software & Data Developer

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github.com/engboda6 engboda6.github.io/portfolio
Born 6/6/2004 · Age 22 · Egypt

Professional Profile

Computer Science student and software developer building end-to-end desktop applications and data analysis systems. Experience spanning JavaScript (React, Electron, Three.js), Python data pipelines, Excel/CSV analytics, and client-ready PDF reporting. Delivers working software with demos, screenshots, and professional documentation—not slides alone.

Education

Computer Science — Faculty of Computers & Information Ongoing

Technical Skills

Programming Languages

- JavaScript (JSX)
- Python
- HTML
- CSS
- SQL
- GLSL
- PowerShell

Frontend & Desktop

- React 18
- Three.js
- Vite
- Electron 33
- electron-builder
- SVG

Backend & APIs

- Node.js
- Express
- REST APIs
- OpenAI SDK
- cors / dotenv

Data & Analytics

- pandas
- numpy
- matplotlib
- openpyxl
- PyMuPDF
- Excel / CSV

Tools & Delivery

- Git / GitHub
- Chrome Headless PDF
- Windows packaging (.exe)
- Client PDF reports

Domain Skills

Desktop app development · 3D visualization · Data cleaning & aggregation · Business loyalty metrics · Technical reporting · Bilingual UI (Arabic / English)

Languages

Arabic — Native · **English** — Professional working proficiency

Experience & Projects

Cosmic Observer — Desktop Astronomy App

2025–2026

Full-Stack Desktop Developer · Solo project

- Built a Windows desktop astronomy observatory with React, Three.js, Node.js/Express, and Electron; shipped NSIS installer and portable .exe.
- Implemented live 3D sky rendering, NEO/debris tracking, sky events, and an optional OpenAI astronomy tutor.
- Integrated real data sources: NASA NeoWs, JPL, CelesTrak TLE, VizieR/CDS; used astronomy-engine for celestial calculations.
- Delivered bilingual Arabic/English UI and published the open-source repository on GitHub.

Vertical Farm Telemetry Analysis

2026

Data Analyst / Python Developer · Solo project

- Designed a pure-Python pipeline over 600 production telemetry rows for a vertical farm.
- Built nutrient-decay analysis across 60 irrigation channels, harvest-weight indices, and optimal feeding balances.
- Produced charts (EC, heatmap, harvest index), Excel exports, and a 3-page client PDF report (Abd Alrahman Ahmed).
- Stack: Python, pandas, numpy, matplotlib, openpyxl.

Home Repair Loyalty Analysis

2026

Data Analyst · Marketing analytics · Solo project

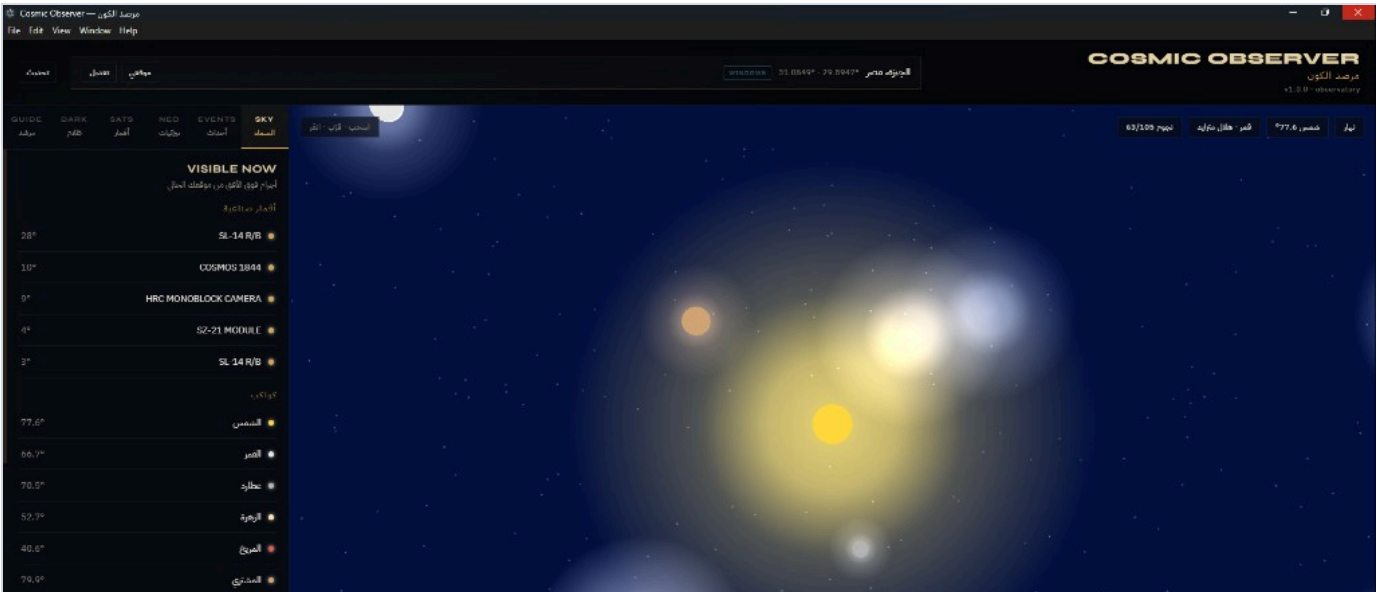
- Analyzed 2,000 marketing records (Jan–Apr 2024) from Data.xlsx for a home-repair business (~\$1.56M revenue, 3.02x ROAS, 49.4% returning customers).
- Measured loyalty by channel, region, and service type; identified Email retention leadership and Referral × Premium as the strongest loyalty pair.
- Recommended post-service surveys and tiered bundles (Care Plus, Premium Shield, Seasonal Autopilot).
- Delivered a dark 5-page client PDF with SVG charts — authored Eng: Abd Alrahman Ahmed.

Core Competencies

End-to-end product delivery · Desktop packaging for Windows · Data cleaning & KPI analysis · API integration · 3D / interactive visualization
· Client reporting (HTML + PDF) · Problem framing from business briefs · Clear technical documentation

Project Highlight — Cosmic Observer

Windows astronomy desktop app: interactive 3D sky, NEO tracking, orbital debris, dark-matter research views, and AI tutor. GitHub: github.com/engboda6/cosmic-observer



Stack used

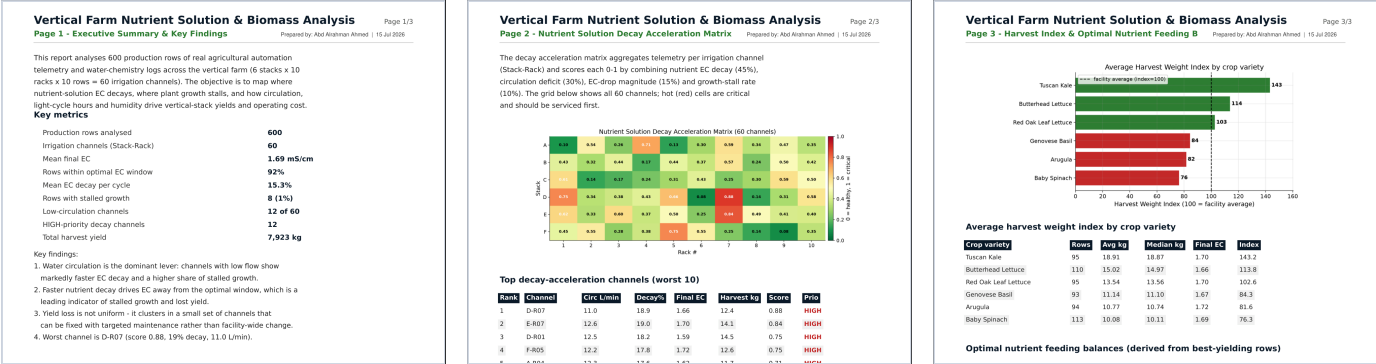
- JavaScript / JSX, HTML, CSS, GLSL, PowerShell
- React 18, Three.js, Vite
- Node.js, Express, astronomy-engine
- OpenAI SDK, Electron 33, electron-builder

Capabilities

- Live 3D sky from GPS / manual location
- NASA / JPL / CelesTrak live feeds
- Sky events, satellites, planets, stars
- Bilingual Arabic + English interface

Project Highlight — Vertical Farm Analysis

Python-only pipeline: raw farm CSV → clean dataset, 60-channel analysis, charts, Excel, and a 3-page PDF prepared by Abd Alrahman Ahmed.



Vertical Farm Nutrient Solution & Biomass Analysis

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Page 3 - Harvest Index & Optimal Nutrient Feeding B

Prepared by: Abd Alrahman Ahmed | 15 Jul 2026

Average Harvest Weight Index by crop variety

Crop variety	Rows	Avg kg	Median kg	Final EC	Index
Tuscan Kale	95	18.91	18.87	1.70	143.2
Butterhead Lettuce	110	15.02	14.97	1.66	113.8
Red Oak Leaf Lettuce	95	13.54	13.56	1.70	102.6
Genovese Basil	93	11.34	11.30	1.67	84.3
Arugula	94	10.77	10.74	1.72	81.6
Baby Spinach	113	10.08	10.11	1.69	76.3

Average harvest weight index by crop variety

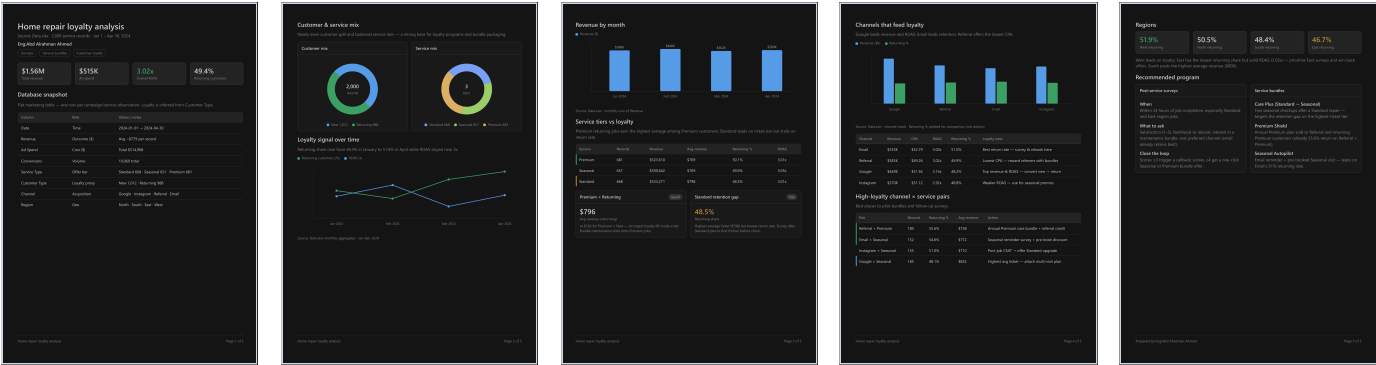
Crop variety	Rows	Avg kg	Median kg	Final EC	Index
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Red Oak Leaf Lettuce	95	13.54	13.56	1.70	102.6
Genovese Basil	93	11.34	11.30	1.67	84.3
Arugula	94	10.77	10.74	1.72	81.6
Baby Spinach	113	10.08	10.11	1.69	76.3

Optimal nutrient feeding balances (derived from best-yielding rows)

Languages / libraries: Python, pandas, numpy, matplotlib, openpyxl

Project Highlight — Home Repair Loyalty

Loyalty analysis from 2,000 marketing rows: surveys + Standard / Seasonal / Premium bundles. Dark 5-page client PDF — Eng: Abd Alrahman Ahmed.



Stack: Python, openpyxl, PyMuPDF, HTML / CSS / SVG, Chrome Headless PDF

Skills Summary by Category

Languages I write in

JavaScript (primary for desktop/web UI) · Python (primary for data & reporting) · HTML / CSS · SQL · GLSL (shaders) · PowerShell (Windows automation)

Frameworks & libraries

React · Three.js · Vite · Electron · Node.js / Express · pandas · numpy · matplotlib · openpyxl · astronomy-engine · OpenAI SDK

What I deliver for clients

Runnable Windows apps · Analysis notebooks/pipelines · Excel outputs · Professional PDF reports · Live demos & screenshots · GitHub repositories · Portfolio documentation

Selected Links

Portfolio website: engboda6.github.io/portfolio

Cosmic Observer repo: github.com/engboda6/cosmic-observer

GitHub profile: github.com/engboda6

Contact

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Available for software development, data analysis, and reporting engagements. Portfolio site includes project pages, screenshots, demo video, and downloadable reports.